

Content and capacitance audit

Reviewed 21 September 2026. Scope: the 16 shipped scenes, their geometry labels, stories, material palette, capacitance calculation, Android About/Help, web/PWA, legacy nanosheet viewer, privacy policy and user-facing documentation.

What has—and has not—been validated

Primary references are catalogued with their supported claims in

[Technical references](#), generated from

`data/references.json`. They are also accessible in Android About and below the web/PWA viewer. The sources support architecture and integration concepts, not the app's exact dimensions, complete material recipe or numerical capacitances.

This is an educational box model, not a foundry PDK, calibrated compact model, process simulation or measured device dataset. A complete physical validation would need doping profiles, material compositions, bias/temperature conditions, realistic dielectric surroundings and a calibrated field/device solver or measurements. Those inputs are absent. No claim of sign-off accuracy is made.

Terminology and story corrections

- **Architecture** is appropriate for FinFET, nanosheet FET, forksheet and CFET. GAA describes gate coverage; nanosheets are one GAA channel shape. CFET describes vertical complementary-device integration, not one compulsory gate connection.
- The displayed path is one possible roadmap. FinFET does not universally stop at a particular node; forksheet is not a mandatory step before CFET. Example-node numbers are illustrative labels, not physical dimensions or process specifications.
- The forksheet model depicts the classic **inner-wall** concept. The newer outer-wall variant is explained but not drawn. The 8nm Si₃N₄ wall is a model choice.
- Monolithic and sequential CFET integration are distinguished without claiming that sequential devices are simply two completed chips glued together. Upper-tier processing after layer transfer must respect the lower tier's thermal budget.
- Common/split gates and frontside/backside contacts are integration options. The depicted backside GND and common inverter input are examples, not CFET definitions.

- Equal fin counts or a geometric p/n width ratio do not establish electrical balance. Mobility, threshold, strain, contacts and operating point also matter. The app displays ideal logic states, not simulated current or timing.
- Rail-to-rail span along z is identified explicitly; standard-cell convention commonly calls this cell height. Gate-stack span and device-pair span are different measurements. Percentage reductions of drawn spans are not equal-drive area gains.
- Layout simplifies the stack and exaggerates vertical dimensions. Its two-sheet nanosheet drawing is not a layer-for-layer rendering of the three-sheet Device/Inverter.
- Perspective camera presets are called **3D overview**, not geometrically exact isometric projections. Legacy nanosheet figure-letter prefixes were removed.

Materials review

Displayed material or role	Interpretation and limits
Si / SiGe	Plausible semiconductor/channel or epitaxy examples. Doping, Ge fraction and strain are unspecified; not every pFET uses the same recipe.
SiO2 + HfO2	Plausible interfacial/high-k stack. Drawn thicknesses are 1nm + 2nm; relative permittivities 3.9 and 22 are model assumptions.
TiN	Conductive work-function region, not “n-doped/p-doped TiN.” Polarity-dependent tuning and interface effects are not modeled.
Mo, Ni, W, NiSi	Illustrative gate/contact palette. Individual uses are plausible; the complete combination is not verified as a manufacturing recipe for these scenes. Chemical labels no longer imply arbitrary routing levels M0–M4.
Si3N4	Illustrative spacer/inner-wall dielectric. Actual spacer and wall compositions are process-dependent.
Middle-tier isolation / bonding oxide	Isolation is a role; MDI chemistry is unspecified. SiO2 bonding dielectric and the assumed MDI permittivity are example choices.
Channel, MD, Po, VD, VG, Metal 0	Layout roles rather than a complete chemical specification. “Po” is a gate-region convention, not a claim of a polysilicon electrode. The internal legacy channel ID nanowire can also draw fins/sheets.

Do the capacitance values make sense?

They are internally checkable **first-order geometry estimates**, with magnitudes consistent with their plate formulas. That does not establish realistic device

capacitance or a reliable architecture ranking. In particular, Cox is an oxide-stack estimate, not terminal Cgg; Cgc/Cge are partial direct couplings, not a full capacitance matrix. Adding these terms does not produce a characterized cell input capacitance.

Corrections

- 1. **Oxide thickness:** volume divided by inner interface area included shell-corner volume and overstated thickness. Use the authored films' actual normal thickness: 1nm SiO2 and 2nm HfO2, rejecting nonuniform-film inputs.
- 2. **Mixed dielectric gaps:** the old midpoint lookup assigned one material to the entire gap. Each sampled ray now integrates thickness/permittivity in series; intervening conductors block that direct ray coupling.
- 3. **Junction selection:** channel-to-body area was counted as source/drain junction area. Only explicitly drawn S/D-to-body contact area now enters the proxy. The FinFET proxy therefore becomes zero because this interface is not drawn, **not because a real FinFET has no junction capacitance.**

Equations and sanity checks

Lengths are nm, areas nm² and capacitances aF (1aF = 10⁻¹⁸ F).

```
epsilon0 = 0.0088541878128 aF/nm
EOT = 1 + 2 × 3.9 / 22 = 1.354545... nm
Cox = epsilon0 × gated_area / (1/3.9 + 2/22)
Cgc, Cge = sum(epsilon0 × sample_area / sum(thickness_i / k_i))
Cj* = epsilon0 × 11.7 × drawn_SD_body_area / assumed_depletion_width
assumed_depletion_width = 5nm
```

Nanosheet: three 30nm × 5nm channels give total gated perimeter

$3 \times 2 \times (30+5) = 210\text{nm}$. At LG=15nm, the area is 3150nm² and

Cox=80.30aF, or 382.4aF/μm of gated perimeter. The previous 358aF/μm was a thickness-extraction artifact, not a physical correction to the plate model.

Scene	Cox before → after (aF)	Cgc before → after (aF)	Cge after (aF)	Cj* after (aF)
FinFET	85.88 → 88.10	15.46 → 15.46	17.61	0.00
Nanosheet	75.12 → 80.30	7.83 → 5.09	16.51	27.35

Scene	Cox before → after (aF)	Cgc before → after (aF)	Cge after (aF)	Cj* after (aF)
Forksheet	107.08 → 112.42	10.60 → 6.95	24.21	40.11
Monolithic CFET	69.83 → 76.48	4.63 → 3.22	12.52	0.00
Sequential CFET	69.83 → 76.48	4.63 → 3.22	12.52	0.00

Identical CFET estimates here reflect the limited geometry estimator, not equivalence of monolithic and sequential technologies. The unchanged Cge numbers do not imply they are experimentally verified.

Remaining assumptions visible in the app

- Unfilled gaps have $k=1$ (vacuum), not a realistic inter-layer dielectric environment.
- Only facing area along x is included for Cgc/Cge. No 3D fringing, corner fields, quantum capacitance, bias-dependent depletion or semiconductor screening is solved.
- S/D epitaxy is approximated as equipotential; doping and bias are unspecified.
- Cj* assumes 5nm depletion and Si permittivity, even where SiGe composition is unspecified. It omits missing junctions and calibrated sidewall behavior.
- W_eff is the total gated perimeter **in that scene**. FinFET/nanosheet show one polarity; forksheet/CFET show both. Normalization is not equal-drive matching.
- Both absolute values **and cross-architecture ratios** depend on these assumptions.

Privacy and provenance

Android saves the guided-tour completion flag locally. Scene controls are not durable profiles, although Android can restore temporary activity state. Android has no INTERNET permission; reference/support links open external apps. Backup is disabled and shared preferences are explicitly excluded in extraction rules.

The website/PWA makes hosting and Google Fonts requests and caches assets. The policy now distinguishes those requests from the native app's behavior; it does not promise that external providers receive no request information.

The old “S. Rathore et al., Semiconductor Science and Technology (2021)” credit lacks a title, DOI and figure number. Its exact source remains **unverified**. It is retained in the reference catalog for provenance, not advertised as a verified reference or silently replaced by a related paper. The original paper/figure is needed to resolve it.

Reproducible checks

From the repository root:

```
python3 -B scripts/build.py
python3 -B scripts/test_parasitics.py
python3 -B scripts/test_content.py
python3 -B scripts/verify.py
python3 scripts/ktcheck.py android/app/src/main/java/io/github/ehsanulkarimpappu/fetlab/*.kt
git diff --check
cd android
./gradlew --offline :app:assembleDebug --no-daemon
```

Capacitance tests include independent gated-perimeter calculations, units, layered dielectrics, mirrored plates, conductor shielding, junction selection and 0.5/0.25/0.125nm grid comparisons (within 3% for these scenes). Content tests check synchronized payloads, reference IDs, model caveats and view naming. Sampled non-overlap checks do not prove electrical connectivity or fabrication feasibility. APK compilation is not real-device testing, and these checks are not TCAD/measurement validation.